ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis — *Expected: December 2026*
B.S. Mechanical Engineering — GPA: 3.65 / 4.00
Relevant Coursework: Numerical Methods & Matrix Algebra, Fluid Mechanics, Heat Transfer, Thermodynamics, Vibrations, Modeling, Simulation, and Control (MATLAB/Simulink), Machine Elements, Solid Mechanics, Machine Shop Practicum

SKILLS

- **Software and Tools:** MATLAB/Simulink, Python (proficient), C++ (working knowledge), PX4, Git, SolidWorks, ANSYS Mechanical, XFLR5, Teensy, Arduino, oscilloscope, multimeter, data acquisition.
- **Technical:** Avionics integration, embedded sensing, sensor calibration, noise characterization, HIL and bench testing, experimental validation, data analysis, CAD, FEA, rapid prototyping, fabrication, technical documentation.

EXPERIENCE

WashU VTOL – *Avionics & Systems Integration Lead*, St. Louis, MO — Sep 2025 – Present
- Integrated avionics and power systems for a prototype eVTOL aircraft including flight controller, GPS, IMU, telemetry radios, and distributed power systems; supported wiring, sensor placement, and system verification.
- Configured motors, ESCs, and PX4 parameters; performed bench tests and HIL checks to validate actuator response and diagnose issues across sensors, onboard computing, and communication systems.
- Supported flight-test setup and post-test review by monitoring system behavior, logging flight data, and documenting integration issues for team review.

Multiplatform Interactive Robotics Lab, UMKC – *Research Intern*, Kansas City, MO — May 2025 – Aug 2025
- Built a Teensy 4.0 sensing platform synchronizing IMU and EMG streams at 1 kHz; developed calibration and noise characterization procedures for repeatable measurements.
- Collected and cleaned synchronized sensor datasets; supported a MATLAB filtering workflow using IMU and vision data for drift-reduced motion tracking.
- Developed Python pipelines for signal conditioning, filtering, timing checks, and validation across motion trials.
- Contributed data collection, experimental methods, and preliminary results for an abstract submission to the IEEE Body Sensor Networks Conference.

WashU Design Build Fly – *Aerodynamics & Payload Engineer*, St. Louis, MO — Sep 2024 – Jan 2026
- Sized carbon fiber wing spars under 2.5g maneuver loads using ANSYS Static Structural FEA, iterating SolidWorks CAD geometry to maintain safety factors with lower mass.
- Designed payload release mechanisms, clamps, and 3D-printed holders with ASME Y14.5 GD&T drawings; fabricated and assembled prototypes with fit checks.
- Conducted XFLR5 aerodynamic trade studies and ANSYS CFD simulations on wing and empennage geometry; compared predictions with flight telemetry.

Washington University McKelvey School of Engineering – *Course Assistant*, St. Louis, MO — Aug 2025 – Dec 2025
- Supported MEMS 2000: Electrical & Electronic Circuits for Mechanical Engineers through lab sessions, weekly office hours, and troubleshooting circuit designs, oscilloscope measurements, and breadboard implementations.
- Graded lab reports and assignments, providing feedback on experimental methodology, data presentation, and technical documentation.

FedEx Corporation – *Material Handler*, Kansas City, MO — Jun 2024 – Aug 2024
- Maintained throughput above 100 packages per hour while following safety protocols in a high-tempo logistics environment.
- Identified loading inefficiencies through cross-team communication and helped reduce handling errors.

TECHNICAL PROJECTS

- **Adaptive Cruise Control System (MATLAB):** Modeled longitudinal vehicle dynamics across three driving regimes and implemented a PID controller, analyzing transient response, solver behavior, and numerical stability.
- **Autonomous Navigation Robot (C++/Arduino):** Built a mobile robot with ultrasonic sensing, H-bridge motor control, and real-time state-machine logic, iterating sensor thresholds and actuator timing through bench testing.
- **Sentinel Prototype (C++ and Python):** Designed an embedded sensing and control prototype with multi-sensor feedback, data acquisition, and closed-loop actuation for autonomous tracking.

INVOLVEMENT

Society of Hispanic Professional Engineers (SHPE) — Sep 2023 – Present
American Society of Mechanical Engineers (ASME) — Aug 2024 – Present
Kessler Scholars Collaborative, Washington University — Jun 2023 – Present